

ATIN SHARMA

Agra, India | atinsharma24@gmail.com | +91 82185 02886
linkedin.com/in/atinsharma24 | github.com/atinsharma24

PROFILE

Backend Engineer specializing in AI-driven data systems, RAG pipeline architecture, and high-performance API development. Expert in Python (Django/FastAPI) and Node.js for building secure, scalable AI backends with advanced retrieval logic and semantic search. Experienced in optimizing database queries and managing large-scale vector embeddings for document intelligence and business automation.

- Production RAG pipeline: sub-800ms retrieval via pgvector + Groq API streaming.
- WhatsApp AI lifecycle: HMAC-SHA256 webhook validation through LLM dispatch to delivery.
- Async document ingestion with 429 retry logic and per-document status tracking.

PROFESSIONAL EXPERIENCE

OpenBiz Software India Pvt Ltd

Jun 2025 – May 2026

Founding Engineer

Remote / Bangalore

- Engineered a non-linear, parallel Graph RAG pipeline using LangGraph state machines to route user queries into asynchronous multi-threaded retrieval paths, reducing end-to-end token latency.
- Implemented an intelligent dual-database ingestion router using Drizzle ORM to stream large-scale business datasets; partitioned unstructured data into Pinecone (Hybrid Dense/Sparse search) while mapping structured compliance data into Supabase (PostgreSQL/pgvector).
- Integrated a real-time web-scraping fallback node using the Massive API headless browser rendering engine and Tavali to inject on-the-fly live Markdown context for time-sensitive regulatory queries.
- Designed runtime evaluation guardrails that intercept Cohere Re-rank v3 cross-encoder scores; implemented an absolute < 0.4 confidence threshold hook that dynamically overrides system prompts to enforce strict context-locking and eliminate LLM hallucinations.
- N+1 fix (JOIN rewrites, composite indexes, TTL cache) — **40% API latency reduction** at 1,000+ user scale.
- CI/CD (GitHub Actions + Vercel): automated testing per PR, staging on merge, prod behind approval — **30% faster deployments**, zero broken-build incidents over four months.

PROJECTS

Nimbus — AI RAG Workspace | github.com/atinsharma24/nimbus

In Development 2026

- Stack: Next.js 14, TypeScript, PostgreSQL, pgvector, Prisma, Groq API, OpenAI
- Async queue ingestion with overlap chunking; pgvector cosine search for hybrid SQL + semantic filtering — sub-800ms retrieval.
- Provider-agnostic LLM layer (groq|openai|anthropic) + SSE streaming; 429 retry with exp. backoff and per-document status tracking.

DocGPT — RAG Document Assistant | github.com/atinsharma24/docgpt

- Stack: React 19, Python (Django), ChromaDB, OpenAI, Docker
- Django RAG pipeline: PyPDF2 overlap chunking, local all-MiniLM-L6-v2 embeddings (offline, zero API cost), ChromaDB cosine retrieval, GPT-4o-mini generation; Dockerized.

Autonomous Web Agent | github.com/atinsharma24/auto-agent

- Stack: Python, Groq, BrowserOS (MCP), Claude 3.5 Sonnet
- Claude 3.5 Sonnet Executor (DOM + form reasoning) + Groq Watcher (page-state validation between steps).
- MCP via BrowserOS for structured DOM access; persistent queue with URL dedup and WAF/CAPTCHA detection.

Automated Content Intelligence Pipeline | github.com/atinsharma24/pipeline

- Stack: React 19, TypeScript, Laravel, OpenAI, Gemini, Docker, Puppeteer
- Puppeteer SPA scraping; Gemini → OpenAI fallback rewriting; React 19 Diff View with per-article state machine; Dockerized microservices on GitHub Actions cron.

SKILLS

Focus Stack: Python, FastAPI, Django, Node.js, PostgreSQL, pgvector, ChromaDB, OpenAI, LangChain, Redis.

Languages: JavaScript/TypeScript, Python, SQL, Java

Frontend: React (v18/19), Next.js, React Native, Redux Toolkit, TailwindCSS

Backend/DB: Node.js, Express, NestJS, Django, PostgreSQL, MongoDB, pgvector, ChromaDB

AI/LLM: OpenAI, Google Gemini, Groq API, RAG Pipelines, LangChain.js, Sentence Transformers, Prompt Engineering

DevOps: Docker, GitHub Actions, Vercel, AWS (S3, DynamoDB)

Tools: Git, Postman, Socket.IO, Razorpay SDK, VideoSDK, Puppeteer

Patterns: Webhook Validation (HMAC-SHA256), N+1 Optimization, Vector Similarity Search, Provider-Agnostic LLM Layers, Idempotent Events

EDUCATION

Vellore Institute of Technology

2026

B.Tech in Computer Science and Engineering

Vellore

- Research (2025): Blockchain-based LLM + FHE for encrypted-data inference without decryption; blockchain provides credential audit trail.